Week 4 Workshop

FIT5230 Malicious AI

Deepfakes I

Overview

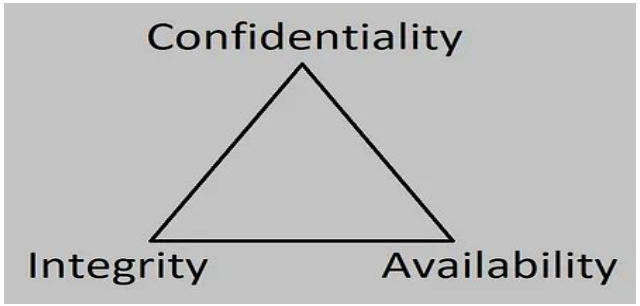
- Refocus on Security Properties

- AI attacks Security
  - CONF
  - INT / AUTH

- Deepfakes
  - 1st order Motion Model
  - Motion-supervised Co-part Segmentation

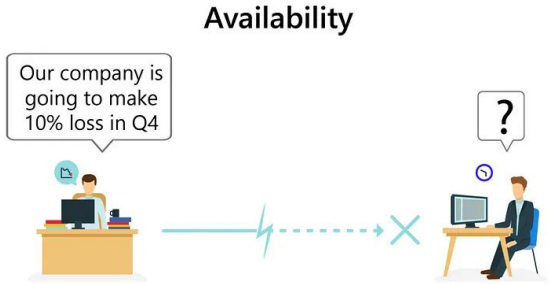
Security Properties - CIA Triad

1. **CONF**identiality: protect information from accidental or intentional disclosure.
2. **INT**egrity: protect information from accidental or intentional (malicious) modification.
3. **AVAIL**ability: Availability means ensuring that the information is available to those who need it and when they need it (to authorized users).









[Image source]

To achieve confidentiality:

- Encryption

To achieve integrity:

- Digital Hash, Digital signature

To achieve availability:

- Load balancing, multi-cloud

4. **AUTH**entication: check if the source/origin is correct
    - check uniqueness of AUTH factor
    - what only you know:
        - passwords, PINs, IC no.
    - what only you have:
        - passport, ATM card
    - what only you are:
        - biometrics
    - static:
        - facial, fingerprint
    - dynamic/soft:
        - gait, gesture, keystrokes
    - who you know:
        - mutual friends

AI attacks Security

- **AI**
  - learn based on past/current observations
  - to recognize/detect: discriminative model
  - to generate/simulate: generative model
- **Security**
  - problems caused by selfish / malicious humans
  - CONFidentiality/INTegrity/ AUTHentication

AI attacks Security

1. **CONF**: inference attacks vs privacy/CONF
   - classify: recognise patterns
   - regress: predict relationships
   - cluster: recognise similar patterns

   Related article:
   https://iapp.org/news/a/privacy-attacks-on-ai-systems-a-current-concern-for-organizations

Inference attack on Database

- Attack:
  - Analyzing functional **dependencies**
  - **Merging** views with same constraints
- Prevent:
  - k-anonymity
  - ℓ-diversity
  - t-closeness
  - differential privacy

Name	Position	Salary ($)	Department	Dep Man.
Andy	Senior		Accounting	Cathy
Chris	Junior		Accounting	Cathy
Cathy	Management		Accounting	Cathy
Julia	Management		R&D	Julia
Maria	Junior		R&D	Julia

Position	Salary ($)
Senior	48,000
Junior	35,000
Management	58,000
Management	62,000
Junior	35,000

Name	Position
Andy	Senior
Chris	Junior
Cathy	Management
Julia	Management
Maria	Junior

Homogeneity attack

Bob	
Zipcode	***Age***
47678	27

Background knowledge attack

Carl	
Zipcode	***Age***
47673	36

A 3-anonymous patient table

Zipcode	Age	Disease
476**	2*	Heart Disease
476**	2*	Heart Disease
476**	2*	Heart Disease
4790*	≥40	Flu
4790*	≥40	Heart Disease
4790*	≥40	Cancer
476**	3*	Heart Disease
476**	3*	Cancer
476**	3*	Cancer

k-anonymity

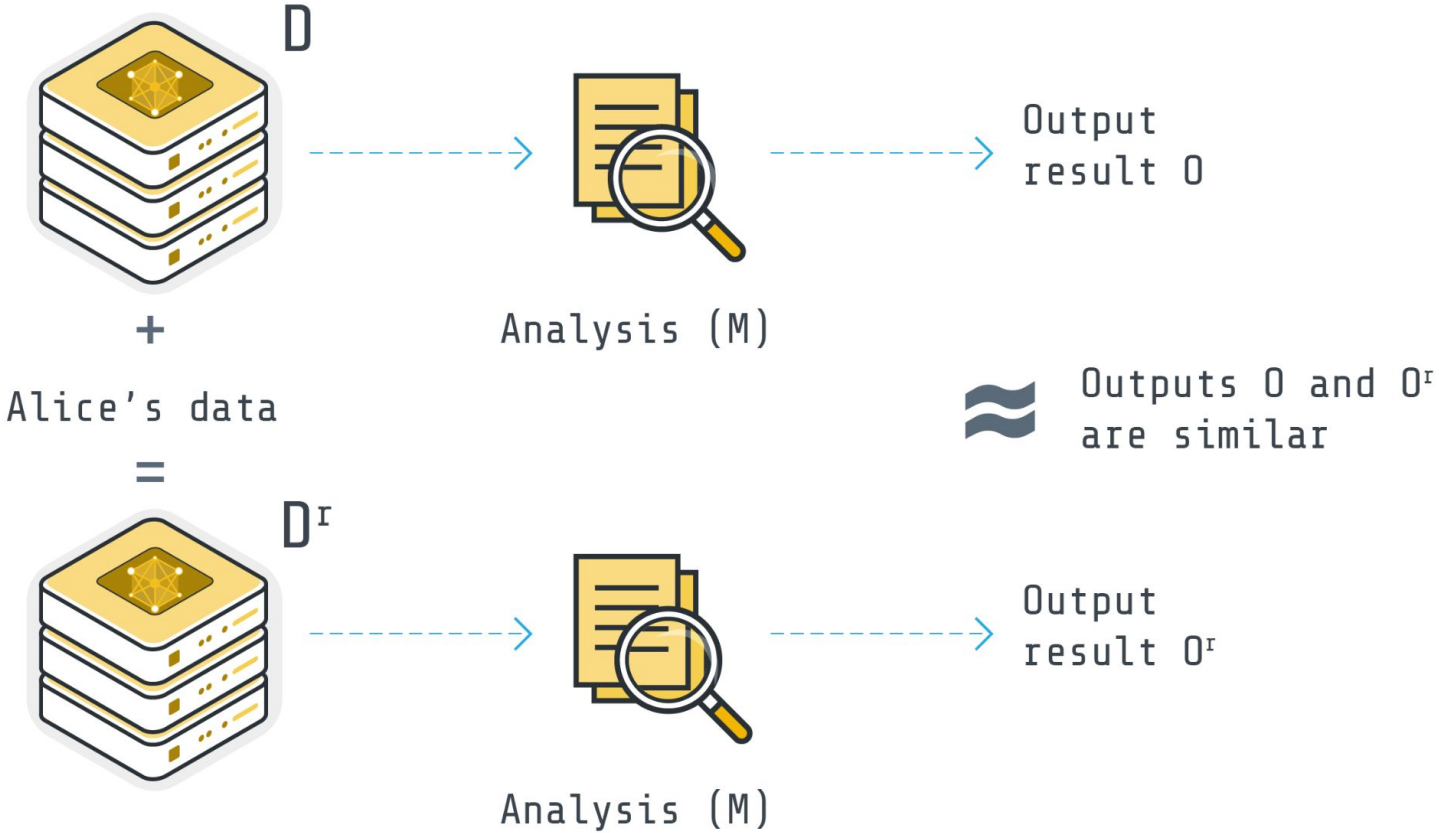
	Nonsensitive			Sensitive
	Zip Code	Age	Nationality	Condition
1	130**	< 30	*	Heart disease
2	130**	< 30	*	Heart disease
3	130**	< 30	*	Viral infection
4	130**	< 30	*	Viral infection
5	1485*	≥ 40	*	Cancer
6	1485*	≥ 40	*	Heart disease
7	1485*	≥ 40	*	Viral infection
8	1485*	≥ 40	*	Viral infection
9	130**	3*	*	Cancer
10	130**	3*	*	Cancer
11	130**	3*	*	Cancer
12	130**	3*	*	Cancer

ℓ-diversity

Non Sensitive		Sensitive	
Age	**Gender**	**Disease**	**Count**
<42	Male	Flu	400
<42	Male	Flu	200
>42	Male	Cancer	400
>42	Female	Cancer	200
>42	Female	Cancer	400

t-closeness

Differential Privacy



https://flower.ai/docs/framework/explanation-differential-privacy.html

2. **INT** & **AUTH**: deepfakes

ARTICLE

Seeing Isn't Believing | Staying Safe During the Deepfake Revolution

BY MEGAN KRUSE 28 MAY 2025

Related article: https://globalcyberalliance.org/about-deepfakes/

First Order Motion Model for Image Animation

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Abstract

Image animation consists of generating a video sequence so that an object in a source image is animated according to the motion of a driving video. Our framework addresses this problem without using any annotation or prior information about the specific object to animate. Once trained on a set of videos depicting objects of the same category (*e.g.* faces, human bodies), our method can be applied to any object of this class. To achieve this, we decouple appearance and motion information using a self-supervised formulation. To support complex motions,

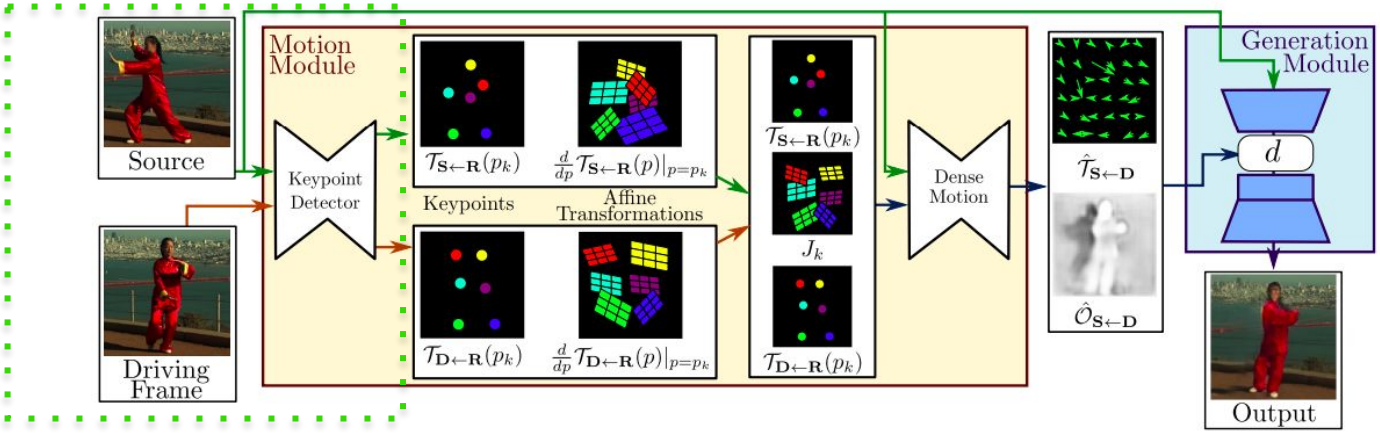
@NIPS 2019

1st Order Motion Model



1st Order Motion Model

- Inputs
  - source image S: extract appearance & retain
  - driving video D: extract motion to animate S
- Output
  - source image S will move like video D

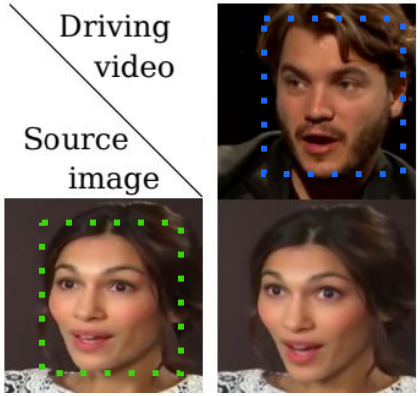


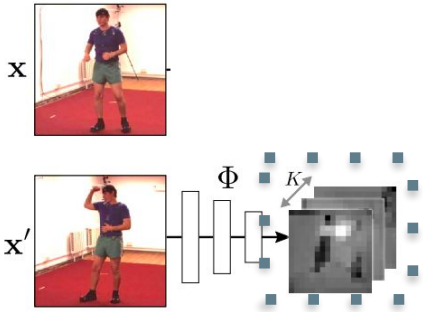
1st Order Motion Model

- Inputs:
  - source image S, driving video D
  - appearance, motion

- Keypoint Detector
  - unsupervised detector
  - predicts key points K from S & D
  - key points / landmarks / heat map of important points

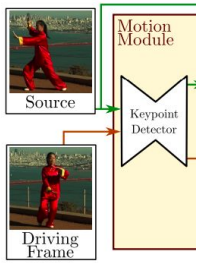


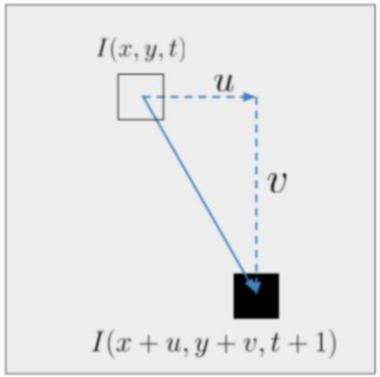




1st Order Motion Model

- Keypoints detected based on
  - comparing two frames (S & D with reference frame)
  - so includes motion info
- motion
  - displacement of pixel location by (u,v): x→x+u , y→y+v
    from current frame t to
    next frame t+1
- I(x,y,t) represents pixel intensity
  at location (x,y) in frame t
- **Q**: I(x+u,y+v,t+1) represents?



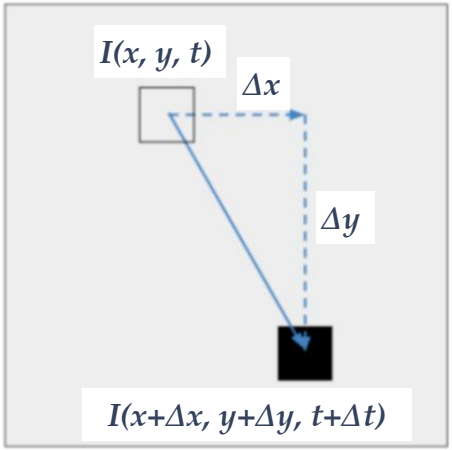




Motion Representation

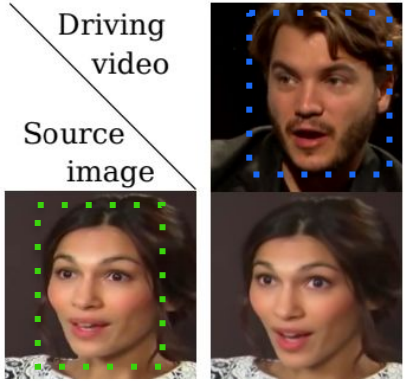
- motion = displacement of pixel I(x,y) by (Δx,Δy) from current frame t to next frame t+Δt
- optical flow (robotics,machine vision)
  - brightness constancy assumption:
    - brightness (intensity) of small patch remains

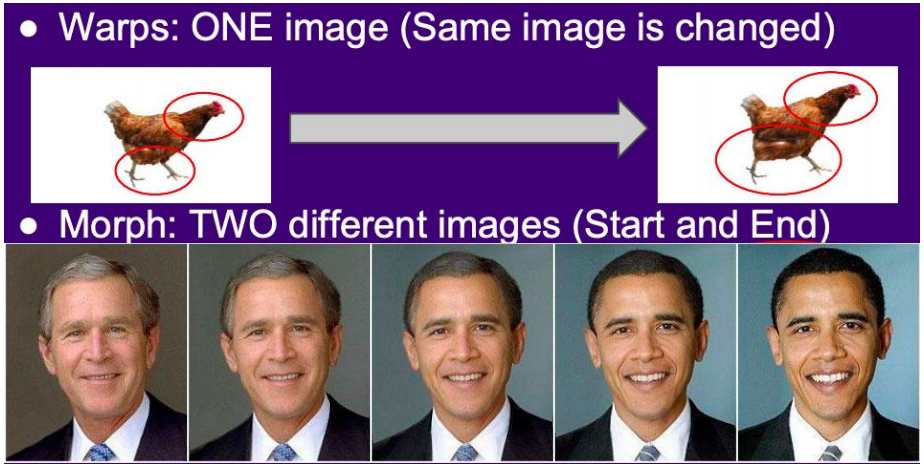
    constant as it moves across time
    - $I(x, y, t) = I(x + \Delta x, y + \Delta y, t + \Delta t)$
  - i.e. though pixel at (x,y) of frame t has moved to (x+Δx, y+Δy) in frame t+Δt, the intensity remains the same



Warping

- Warping
  - change/distort the form/shape
- Morphing
  - gradual transformation of one image to another





https://www.giss.nasa.gov/edu/nycri/research/files/2015/2015-Hostos-Prince_presentation.pdf

Motion-supervised Co-Part Segmentation

Aliaksandr Siarohin[1*], Subhankar Roy[1,4*], Stéphane Lathuilière[2], Sergey Tulyakov[3], Elisa Ricci[1,4], and Nicu Sebe[1,5]

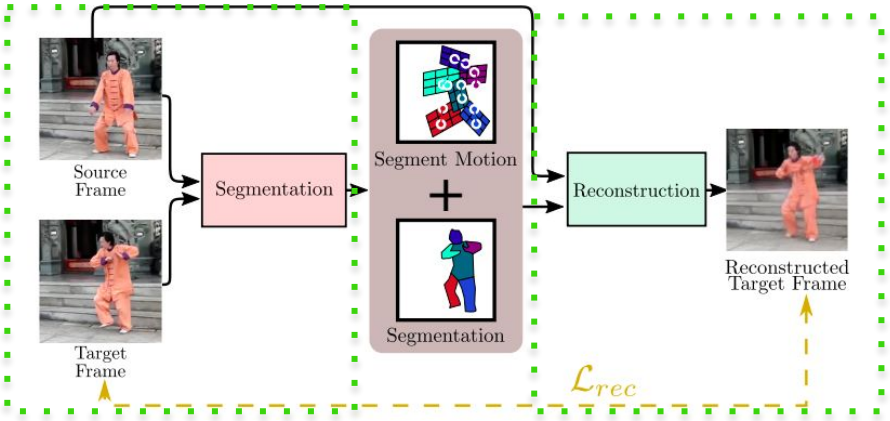
[1]DISI, University of Trento; [2]LTCI, Tlcom Paris, Institut polytechnique de Paris;
[3]Snap Inc.; [4]Fondazione Bruno Kessler; [5]Huawei Technologies Ireland

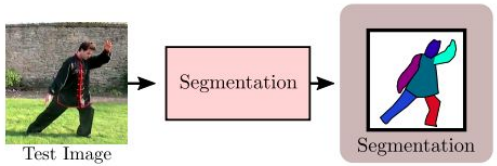
Abstract. Recent co-part segmentation methods mostly operate in a supervised learning setting, which requires a large amount of annotated data for training. To overcome this limitation, we propose a self-supervised deep learning method for co-part segmentation. Differently from previous works, our approach develops the idea that motion information inferred from videos can be leveraged to discover meaningful object parts. To this end, our method relies on pairs of frames sampled from the same video. The network learns to predict part segments together with a representation of the motion between two frames, which permits reconstruction of the target image. Through extensive experi-

@ICPR 2021

Motion-based CoSegmentation

- Inputs
  - videos D having objects of same class w diff appearance
  - source image S
- Output
  - segment from image S transferred to video D





Motion-based CoSegmentation

- Inputs: video D , source image S
- Output:
  - segment from image S transferred to video D





Motion-based CoSegmentation

- Training: 2 random frames from input video D

- Segmentation
  - segment into K+1 parts (K for foreground, 1 background)
  - each segment groups pixels that move together based on a segment-wise optical flow
- Reconstruction
  - reconstruct target frames of D:
    - warp frame based on optical flow from S to D

Activity 1

Identifying Deepfakes

Go through the list of images, could you tell which one of them is real or fake?

Image 1



Image 2



Image 3



Image 4



Image 5



Image 6



Image 7



Image 8:



Image 9:



Image 10



What you need to do:

Given these scenario:

1. Political Deepfake: A fake video of a politician making false claims goes viral before an election.
2. Corporate Espionage: A deepfake voice call impersonates a CEO to authorize a fraudulent transaction.
3. Social Media Manipulation: A celebrity's face is used in fake videos for clickbait or misinformation.
4. Medical Misinformation: A deepfake video spreads false health advice using a trusted doctor's likeness.

Discuss the following:

1. What are the security properties (Confidentiality, Integrity, Availability, Authentication) being violated?
2. How could AI models like the 1st Order Motion Model be used to create or detect this deepfake?
3. What technical or policy-based defenses could be implemented?
4. Who should be held accountable?

Provide your answer here: https://shorturl.at/FuxxB